Shubham Thakur

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LinkedIn GitHub Personal Site

Summary

Software Engineer with **2+ years of industry experience** across internships building full-stack applications, AI-powered tools, and developer solutions. Proficient in TypeScript, Python, and Scala, with interests in intelligent systems, developer tools, and distributed systems. I have contributed to enterprise teams at **Morgan Stanley** and **ServiceNow**, developing tools that simplify complex workflows. I want to further strengthen my engineering skills, learn from experienced teams, and contribute to high-impact software products.

Experience

ServiceNow **May 2026 – Jul 2026**

Associate Software Engineer - Intern *Hyderabad, India*

- Created an AI-powered JavaScript dashboard unifying data from **3+ ServiceNow portals**, enabling users to build custom data cards and generate AI-driven summaries for developer- and manager-specific views.
- Expanded end-to-end test coverage for App Engine Studio by developing **25+ automated tests** using **Java** and **Playwright**.

Morgan Stanley **Jul 2024 – Apr 2026**

Technology Apprentice *Bengaluru, India*

- Built a **Scala** tool for the Counterparty Valuation Adjustment team to manage **UNIX environment overrides** and **Symphony parameters**, replacing a CLI-intensive workflow with an interactive UI and audit trail that **reduced manual effort by nearly 50%**.
- Developed an upstream data anomaly detection PoC using **Snowflake ML** and the **Isolation Forest** algorithm to automate anomalous data identification.
- Prototyped an **agentic issue-resolution chatbot** using the **OpenAI SDK** and **OpenAI Embeddings** to streamline incident investigation and resolution.

Projects

Renaissance | *Electron, React, TypeScript, Fastify, Supabase, Git, pnpm*

github.com/suu-b/renaissance

- Developing a **local-first desktop application** using **Electron, React, and TypeScript** for offline writing and collaboration.
- Designed a **3-tier architecture** connecting a local Node.js service with a remote **Fastify backend**.
- Abstracted Git workflows into a custom service layer with **commit history** and multi-user collaboration support.
- Implemented **OAuth 2.0 with PKCE**, deep-link callbacks, and automatic token refresh.
- Secured credentials using **keytar** and enforced database access with Supabase **10+ Row-Level Security Policies**.
- Developed **30+ type-safe REST API endpoints** using **Fastify, Zod, and TypeScript**.

Event Bus | *Python, FastAPI, Redis, Multiprocessing, SSE*

github.com/suu-b/event-bus

- Created a **multiprocessing-based request dispatch engine** with competing worker processes and synchronization primitives.
- Migrated to a **Redis-backed distributed architecture** using Pub/Sub for worker communication.
- Implemented atomic **SET NX claims** to enforce exclusive request ownership and prevent duplicate processing.
- Added **real-time progress streaming** using **FastAPI SSE** and Redis.
- Benchmarked **10–100 concurrent requests** with **10 workers**, measuring end-to-end latency and queue wait.
- Observed mean queue wait increase from **1.74s to 50.83s** as concurrent requests increased from 10 to 100.

Education

Chitkara University **Jul 2023 – Jul 2027**

Bachelor of Engineering in Computer Science | CGPA: 10.0 *Himachal Pradesh, India*

Technical Skills

Languages: Python, TypeScript, JavaScript, Java, C++, Scala, SQL, HTML/CSS
Frameworks: React, Electron, Fastify, FastAPI, Node.js, Tailwind CSS, Pydantic, Zod
AI/ML: OpenAI SDK, OpenAI Embeddings, RAG, Snowflake ML, Isolation Forest
Databases: PostgreSQL, Supabase, Redis
Infrastructure & Developer Tools: Git, GitHub, Linux, Playwright, pnpm, OpenAPI